

Qualification Pack



AI DevOps Analyst

QP Code: SSC/Q8112

Version: 4.0

NSQF Level: 5

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SSC/Q8112: AI DevOps Analyst

Brief Job Description

A DevOps Engineer is a key player in the software development lifecycle, responsible for bridging the gap between development, operations, and quality assurance teams. They design and maintain Continuous Integration and Continuous Delivery (CI/CD) pipelines to automate testing, building, and deployment processes, ensuring high-quality software delivery at speed. Utilizing tools like Terraform and Ansible, they automate the provisioning and configuration of infrastructure across various environments, while also implementing monitoring and logging solutions to optimize application performance. DevOps Engineers incorporate security practices (DevSecOps) into the development lifecycle, manage incident response strategies, and emphasize collaboration and documentation to foster continuous improvement within the organization. Through these efforts, they ensure that applications are robust, scalable, and aligned with business objectives..

Personal Attributes

An AI - Devops Engineer needs to be well organized and have excellent communication skills to be a bridge between siloed teams and work towards a common goal. S/he also needs to be flexible to work for long hours to ensure continuity in operations.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8120: Develop tools, processes and mechanisms for continuous integration and delivery](#)
2. [SSC/N8155: Develop and maintain CI/CD Pipelines](#)
3. [SSC/N8156: Infrastructure management for training & deploying AI models](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
Country	India
NSQF Level	5

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Credits	16
Aligned to NCO/ISCO/ISIC Code	NCO-2015/251.NIL
Minimum Educational Qualification & Experience	Completed 3-year diploma (after 10th) (PG/UG/diploma with courses related to Engg./ Science) with 1.5 years of experience in relevant field.Relevant Experience: AI/Computer Science/IT The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualifications. OR Previous relevant Qualification of NSQF Level (4.5) with 3 Years of experience in relevant field.Relevant Experience: AI/Computer Science/IT The relevant experience would include work, internship and apprenticeship undertaken post-completion of relevant educational qualifications OR Completed 2nd year of UG (UG Diploma) (diploma PG/UG/diploma with courses related to Engg./ Science)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-05-IT-045822025-V2-NASSCOM
NQR Version	2

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SSC/N8120: Develop tools, processes and mechanisms for continuous integration and delivery

Description

This unit is about managing the administration of production systems and operations to assist continuous delivery and integration.

Scope

The scope covers the following :

- Deploy production systems
- Monitor performance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Grasp the high-level goals of the project, including desired outcomes, customer needs, and success criteria
- PC2.** Identify key stakeholders and their expectations to ensure the project aligns with business objectives and customer requirements
- PC3.** Understand the software development methodology being used and how it impacts collaborative practices and project timelines
- PC4.** Familiarize with version control systems and the established workflow for code collaboration
- PC5.** Comprehend the pipeline setup for continuous integration and continuous delivery
- PC6.** Understand the testing strategies that are integrated into the CI/CD pipeline to ensure quality.
- PC7.** Assess the infrastructure requirements for the application, including scaling considerations, cloud services, and resource allocation.
- PC8.** Familiarity with tools and practices that ensure the infrastructure is maintained consistently across different environments
- PC9.** Identify the metrics that will be monitored to measure application performance, system health, and user experience in production
- PC10.** Understand how logging will be implemented for troubleshooting and performance analysis, including the tools used for log aggregation and analysis
- PC11.** Recognize the importance of integrating security measures within the development and deployment processes to mitigate risks
- PC12.** Be aware of any regulatory or compliance requirements that the software must adhere to, which may influence design and operational practices
- PC13.** Understand how user feedback will be collected and used to inform development, identify issues, and enhance product features
- PC14.** A clear understanding of how incidents will be handled, including procedures for diagnosing issues, communication protocols, and recovery strategies

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PC15. Knowledge of disaster recovery strategies to ensure business continuity in case of critical system failures.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to devops
- KU2.** organizational policies and procedures for sharing data
- KU3.** who to involve when managing administration of production systems and operations
- KU4.** Project Management Principles: Understanding high-level goals, customer needs, and identifying success criteria to align project outputs with business objectives.
- KU5.** Stakeholder Engagement: Familiarity with stakeholder identification methods and techniques to gather and manage their expectations and requirements.
- KU6.** Software Development Methodologies: Knowledge of various methodologies (e.g., Agile, Scrum, Kanban) and their principles to drive collaborative practices and manage timelines effectively.
- KU7.** Version Control Systems: Proficiency in tools like Git for source code management, including an understanding of branching strategies, merging processes, and pull requests to ensure collaborative code development.
- KU8.** CI/CD Pipeline Fundamentals: In-depth knowledge of how to set up and maintain CI/CD pipelines, focusing on test automation, build automation, and automated deployment processes.
- KU9.** Testing Strategies: Understanding different types of testing (unit tests, integration tests, end-to-end tests) and their roles within the CI/CD pipeline to ensure software quality and reliability.
- KU10.** Infrastructure Assessment: Familiarity with infrastructure requirements, including scaling considerations, cloud services (e.g., AWS, Azure, Google Cloud), and resource allocation for applications.
- KU11.** Infrastructure Consistency: Knowledge of tools and practices for maintaining infrastructure consistency across environments (e.g., Infrastructure as Code (IaC) tools like Terraform, Ansible, or Docker).
- KU12.** Performance Metrics: Ability to identify and define relevant metrics to monitor application performance, system health, and user experience in production environments.
- KU13.** Logging and Monitoring: Understanding of logging practices and tools (e.g., ELK Stack, Splunk) for troubleshooting and performance analysis, as well as log aggregation methods.
- KU14.** Security in CI/CD: Awareness of integrating security measures (DevSecOps principles) into CI/CD processes to address vulnerabilities and mitigate risks associated with development and deployment.
- KU15.** Regulatory Compliance: Knowledge of relevant regulatory or compliance requirements (e.g., GDPR, HIPAA) and their influence on software design, development, and operational practices.
- KU16.** User Feedback Mechanisms: Familiarity with approaches for gathering user feedback, tracking issues, and utilizing insights to drive product enhancements and feature development.

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- KU17.** Incident Management: Understanding of incident management protocols, including issue diagnosis, communication strategies, and recovery plans to handle incidents effectively.
- KU18.** Disaster Recovery: Knowledge of disaster recovery planning and strategies to ensure application availability and business continuity following critical system failures.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable courses
- GS2.** plan and organize your own work to achieve targets and deadlines
- GS3.** refer anomalies to the supervisor
- GS4.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Grasp the high-level goals of the project, including desired outcomes, customer needs, and success criteria	2	3	-	1
PC2. Identify key stakeholders and their expectations to ensure the project aligns with business objectives and customer requirements	2	3	-	1
PC3. Understand the software development methodology being used and how it impacts collaborative practices and project timelines	2	4	-	2
PC4. Familiarize with version control systems and the established workflow for code collaboration	2	4	-	2
PC5. Comprehend the pipeline setup for continuous integration and continuous delivery	2	4	-	2
PC6. Understand the testing strategies that are integrated into the CI/CD pipeline to ensure quality.	2	4	-	2
PC7. Assess the infrastructure requirements for the application, including scaling considerations, cloud services, and resource allocation.	2	4	-	2
PC8. Familiarity with tools and practices that ensure the infrastructure is maintained consistently across different environments	2	3	-	1
PC9. Identify the metrics that will be monitored to measure application performance, system health, and user experience in production	2	3	-	1
PC10. Understand how logging will be implemented for troubleshooting and performance analysis, including the tools used for log aggregation and analysis	2	3	-	1
PC11. Recognize the importance of integrating security measures within the development and deployment processes to mitigate risks	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. Be aware of any regulatory or compliance requirements that the software must adhere to, which may influence design and operational practices	2	3	-	1
PC13. Understand how user feedback will be collected and used to inform development, identify issues, and enhance product features	2	3	-	1
PC14. A clear understanding of how incidents will be handled, including procedures for diagnosing issues, communication protocols, and recovery strategies	2	3	-	1
PC15. Knowledge of disaster recovery strategies to ensure business continuity in case of critical system failures.	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8120
NOS Name	Develop tools, processes and mechanisms for continuous integration and delivery
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics, Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	4
Version	5.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8155: Develop and maintain CI/CD Pipelines

Description

Developing and maintaining Continuous Integration/Continuous Deployment (CI/CD) pipelines involves several key activities that ensure smooth, automated, and reliable software delivery.

Scope

The scope covers the following :

- Collaboration and Communication, Pipeline design and setup, Version Control Management, Build automation, Automated testing, Deployment automation, Monitoring and logging, Pipeline Maintenance and Optimization, Documentation, Security Considerations, Feedback Loop and Reporting

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Define continuous delivery and integration strategies and participate in Agile ceremonies to align team efforts and adjust workflows accordingly
- PC2.** Outline key stages of the pipeline such as build test deploy and release and select appropriate tools to support each stage based on project needs
- PC3.** Configure triggers webhooks and repository structures to automate pipeline execution upon code changes and facilitate collaboration using branching strategies
- PC4.** Write build scripts and set up artifact repositories to compile source code generate build artifacts and store them for future use
- PC5.** Automate testing at various levels including unit integration and end-to-end functional tests and integrate them into the pipeline
- PC6.** Manage environment configurations and scripts for automated deployment
- PC7.** Integrate monitoring tools and logging frameworks to track application performance and enable troubleshooting
- PC8.** Keep CI/CD tools dependencies and security scanning up to date and employ secret management solutions to handle sensitive data securely
- PC9.** establish notifications for team alerts and collect metrics such as build success rates deployment frequency and test coverage for ongoing process improvement
- PC10.** Document pipeline workflows configurations and troubleshooting procedures to ensure maintainability and facilitate knowledge sharing

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to devops
- KU2.** organizational policies and procedures for sharing data
- KU3.** who to involve when managing administration of production systems and operations

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- KU4.** Understanding high level project goals customer needs and success criteria to align outputs with business objectives
- KU5.** Familiarity with stakeholder identification techniques managing expectations and requirements
- KU6.** Knowledge of software development methodologies version control systems and CI/CD pipeline fundamentals for effective collaboration and automation
- KU7.** Awareness of infrastructure assessment infrastructure consistency performance metrics logging monitoring security in CI/CD and regulatory compliance for reliable and secure systems
- KU8.** Understanding user feedback incident management and disaster recovery strategies to enhance product quality and ensure business continuity
- KU9.** Establish notifications for team alerts and collect metrics such as build success rates deployment frequency and test coverage for ongoing process improvement

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** plan and organize your own work to achieve targets and deadlines
- GS2.** refer anomalies to the supervisor
- GS3.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Define continuous delivery and integration strategies and participate in Agile ceremonies to align team efforts and adjust workflows accordingly	3	5	-	2
PC2. Outline key stages of the pipeline such as build test deploy and release and select appropriate tools to support each stage based on project needs	3	5	-	2
PC3. Configure triggers webhooks and repository structures to automate pipeline execution upon code changes and facilitate collaboration using branching strategies	3	5	-	2
PC4. Write build scripts and set up artifact repositories to compile source code generate build artifacts and store them for future use	3	5	-	2
PC5. Automate testing at various levels including unit integration and end-to-end functional tests and integrate them into the pipeline	3	3	-	2
PC6. Manage environment configurations and scripts for automated deployment	3	5	-	2
PC7. Integrate monitoring tools and logging frameworks to track application performance and enable troubleshooting	3	5	-	2
PC8. Keep CI/CD tools dependencies and security scanning up to date and employ secret management solutions to handle sensitive data securely	3	5	-	2
PC9. establish notifications for team alerts and collect metrics such as build success rates deployment frequency and test coverage for ongoing process improvement	3	5	-	2
PC10. Document pipeline workflows configurations and troubleshooting procedures to ensure maintainability and facilitate knowledge sharing	3	5	-	2

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8155
NOS Name	Develop and maintain CI/CD Pipelines
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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SSC/N8156: Infrastructure management for training & deploying AI models

Description

In context of DevOps, infrastructure management encompasses a range of activities focused on ensuring that the underlying infrastructure supports continuous development, testing, and deployment processes efficiently and reliably

Scope

The scope covers the following :

- Infrastructure as Code (IaC), Configuration Management, Continuous Monitoring, Automated Scaling and Load Balancing, Backup and Disaster Recovery, Security Management, Incident Management and Troubleshooting, Patch Management and Updates, Documentation and Knowledge Sharing, Continuous Improvement

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Use tools to automate the provisioning of infrastructure components enabling consistent environments across development.
- PC2.** Store infrastructure code in version control systems to track changes over time and enable collaboration among teams
- PC3.** Utilize configuration management tools to automatically configure servers and applications to maintain consistent states
- PC4.** Ensure that all environments are configured according to the same standards to minimize discrepancies
- PC5.** Implement monitoring tools to collect metrics on infrastructure performance and resource utilization
- PC6.** Monitor application performance to identify bottlenecks or issues related to infrastructure
- PC7.** Implement auto scaling solutions to automatically adjust resources based on traffic or workload demands
- PC8.** Use load balancers to distribute incoming traffic across multiple servers
- PC9.** Set up automated backup solutions for critical data and applications to ensure recovery options are readily available.
- PC10.** Develop and test disaster recovery plans including failover strategies to ensure business continuity
- PC11.** Implement security practices by incorporating security scans and compliance checks into CI/CD pipelines.
- PC12.** apply role-based access controls and regularly audit permissions to maintain a secure infrastructure
- PC13.** Create and implement plans for responding to infrastructure incidents for diagnosis and resolution

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- PC14.** Conduct thorough analyses of incidents to identify and address underlying issues preventing future occurrences
- PC15.** Schedule and automate updates to keep the infrastructure secure and upto date
- PC16.** Before deploying updates, test them in staging environments to mitigate risks associated with changes
- PC17.** Maintain comprehensive and up-to-date documentation
- PC18.** Encourage collaboration and knowledge sharing within teams to foster a culture of learning and adaptability
- PC19.** Establish mechanisms for collecting feedback on infrastructure performance and areas for improvement using insights to refine processes
- PC20.** Regularly evaluate and adopt new technologies and tools that can enhance infrastructure efficiency and support evolving business needs

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies, procedures and guidelines which relate to devops
- KU2.** organizational policies and procedures for sharing data
- KU3.** who to involve when managing administration of production systems and operations
- KU4.** Understanding various continuous delivery and deployment models and designing staging environments effectively
- KU5.** Familiarity with infrastructure as code processes and configuration management techniques
- KU6.** Knowledge of continuous integration and build automation methods and application release automation approach
- KU7.** Awareness of application lifecycle management practices and testing methodologies such as unit and integration testing
- KU8.** Experience working across different operating systems and utilizing test automation tools

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** plan and organize your own work to achieve targets and deadlines
- GS2.** refer anomalies to the supervisor
- GS3.** check your work is complete and free from errors

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Use tools to automate the provisioning of infrastructure components enabling consistent environments across development.	1	2	-	1
PC2. Store infrastructure code in version control systems to track changes over time and enable collaboration among teams	1	2	-	1
PC3. Utilize configuration management tools to automatically configure servers and applications to maintain consistent states	1	2	-	1
PC4. Ensure that all environments are configured according to the same standards to minimize discrepancies	1	2	-	1
PC5. Implement monitoring tools to collect metrics on infrastructure performance and resource utilization	1	2	-	1
PC6. Monitor application performance to identify bottlenecks or issues related to infrastructure	1	2	-	1
PC7. Implement auto scaling solutions to automatically adjust resources based on traffic or workload demands	1	2	-	1
PC8. Use load balancers to distribute incoming traffic across multiple servers	1	2	-	1
PC9. Set up automated backup solutions for critical data and applications to ensure recovery options are readily available.	1	2	-	1
PC10. Develop and test disaster recovery plans including failover strategies to ensure business continuity	1	2	-	1
PC11. Implement security practices by incorporating security scans and compliance checks into CI/CD pipelines.	2	3	-	1

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. apply role-based access controls and regularly audit permissions to maintain a secure infrastructure	2	3	-	1
PC13. Create and implement plans for responding to infrastructure incidents for diagnosis and resolution	2	3	-	1
PC14. Conduct thorough analyses of incidents to identify and address underlying issues preventing future occurrences	2	3	-	1
PC15. Schedule and automate updates to keep the infrastructure secure and upto date	2	3	-	1
PC16. Before deploying updates, test them in staging environments to mitigate risks associated with changes	2	3	-	1
PC17. Maintain comprehensive and up-to-date documentation	2	3	-	1
PC18. Encourage collaboration and knowledge sharing within teams to foster a culture of learning and adaptability	2	3	-	1
PC19. Establish mechanisms for collecting feedback on infrastructure performance and areas for improvement using insights to refine processes	2	3	-	1
PC20. Regularly evaluate and adopt new technologies and tools that can enhance infrastructure efficiency and support evolving business needs	2	3	-	1
NOS Total	30	50	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8156
NOS Name	Infrastructure management for training & deploying AI models
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

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DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts.

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8120.Develop tools, processes and mechanisms for continuous integration and delivery	30	50	-	20	100	30
SSC/N8155.Develop and maintain CI/CD Pipelines	30	50	-	20	100	30
SSC/N8156.Infrastructure management for training & deploying AI models	30	50	-	20	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.